TP 6:Optimisation PSO

Let's consider the following algorithm PSO

```
initialise values of ω, φp, φg(proposition: 0..1)
initialize values of blo and bup
initialise values of lr0 ≤ lr ≤ 1.0
initialise values of rp and rg
Initialize g~ U(blo, bup), compute g
for each particle i = 1, ..., Sdo
    Initialize the particle's position with a uniformly distributed random vector: xi ~ U(blo, bup)
    Initialize the particle's best-known position to its initial position: pi ← xi
Compute f(xi) (or f(pi))
Iff(pi) <f(g) then
        update the swarm's best known position: g ← pi
    Initialize the particle's velocity: vi ~ U(-|bup-blo|, |bup-blo|)
##########################################################################################
while a termination criterion is not met do:/// use the number of iterations
for each particle i = 1, ..., Sdo
for each dimension d = 1, ..., ndo// only if you have multidimensional variables
            Pick random numbers: rp, rg ~ U(0,1)
            Update the particle's velocity: vi,d ← ωvi,d + φprp (pi,d-xi,d) + φgrg (gd-xi,d)
Update the particle's position: xi ← xi + lr vi
iff(xi) <f(pi) then
Update the particle's best known position: pi ← xi
iff(pi) <f(g) then
Update the swarm's best known position: g ← pi

while a termination criterion is not met do:/// use the number of iterations
for each particle i = 1, ..., Sdo
            Pick random numbers: rp, rg ~ U(0,1)
            Update the particle's velocity: vi ← ωvi + φprp (pi-xi) + φgrg (g-xi)
Update the particle's position: xi ← xi + lr vi
Iff(xi) <f(pi) then
Update the particle's best known position: pi ← xi
Iff(pi) <f(g) then
Update the swarm's best known position: g ← pi



Print (g)
```

Questions:

1. Propose an implementation of PSO to minimise the function
   f(x)=10*sin((0.3*x)*sin(1.3*(x^2)+0.00001*(x^4)+0.2*x+80)).

- The values $\mathbf{b}_{lo}$ and $\mathbf{b}_{up}$ represent the lower and upper boundaries of the search-space respectively.
- The termination criterion can be the number of iterations performed, or a solution where the adequate objective function value is found.[11]
- The parameters ω, φ_p, and φ_g are selected by the practitioner and control the behaviour and efficacy of the PSO method.
- **lr** represents the learning rate (0 ≤ **lr** ≤ 1.0), which is the proportion at which the velocity affects the movement of the particle (where **lr** = 0 means the velocity will not affect the particle at all and **lr** = 1 means the velocity will fully affect the particle).